

TOKEN BALANCING, UPDATE SPARSITY, AND SUPERVISION DENSITY IN MULTI-TEACHER ON-POLICY DISTILLATION

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ABSTRACT

Multi-teacher on-policy distillation (MOPD) is an emerging approach for consolidating specialized teacher capabilities into a shared student; yet, how different training designs influence the integration of teacher capabilities and student performance remains unexplored. In this paper, we investigate three questions in MOPD, namely loss averaging, parameter-update sparsity, and vocabulary supervision, using 1.7B and 3B LLMs on mathematics, coding, question answering, and instruction-following tasks. Comparing global-token (GT), domain-token (DT), and domain-response (DR) loss averaging reveals a trade-off between average performance and capability retention. GT achieves the highest mean score across domains, while DR produces the smallest decline from base model in its most affected domain. Measurements during joint training show that after 250 updates, only 6–7% of BF16 model parameters differ from initialization, compared with approximately 96% of FP32 master parameters. Controlled local BF16 updates from a shared student and Adam state show mean pairwise cosine similarity of about 0.85 between teachers and overlap on about 70% of the coordinates significantly changed by either teacher. Vocabulary supervision specifies how many top-ranked tokens’ teacher logits enter the loss. At a common student and optimizer state, the compared supervision settings yield similar local update sparsity and concentration. In joint training at update 250, top- k supervision achieves higher mathematics, coding, and question-answering scores than sampled-token supervision, while matching instruction following; single-teacher training retains cross-domain trade-offs. Together, these results show that supervision balance and update concentration alone do not characterize successful capability consolidation, motivating direct evaluation of per-domain capability trajectories. Building on these findings, we propose a training recipe combining explicit supervision weights, capability-based selection of vocabulary feedback, and checkpoint selection under per-domain retention constraints, with its stability left to further validation.

1 INTRODUCTION

A language model may need mathematical reasoning, code generation, scientific knowledge, and instruction following within the same deployment. Multi-teacher on-policy distillation (MOPD) transfers such capabilities by having specialized teachers score responses generated by a shared student (Ma et al., 2026). Because the student supplies the responses, its current behavior determines both the examples on which teachers provide feedback and the lengths over which that feedback is averaged.

Equal numbers of prompts from different domains need not give those domains equal weight in the loss. Open-MOPD identifies response-length imbalance as a source of unequal supervision (Gao et al., 2026). This raises a more specific question about the objective: does an averaging rule balance domains, responses within each domain, or both? These choices can be separated algebraically and compared empirically through the capabilities of the resulting student.

The parameter changes induced by teacher feedback raise a second question. Prior work finds concentrated changes in reinforcement learning (Mukherjee et al., 2025) and distillation (Yu et al., 2026; Shen et al., 2026). In a shared student, each teacher acts on parameters already influenced by other teachers. Concentration alone does not specify whether teachers affect the same locations. The relevant comparison therefore begins with a common student and optimizer state, and relates teacher-specific parameter changes to differences between teacher predictions.

A third question concerns the breadth of teacher feedback. A sampled-token loss and a loss over several vocabulary alternatives supervise the same generated prefix differently. Their vocabulary coverage does not directly determine how many model parameters change: the softmax couples logits, and the optimizer combines the current gradient with its history. Comparing both update concentration and capability connects the local learning signal to training outcomes.

We organize the paper around these three questions:

1. **Token balancing (Section 3).** We derive how global-token, domain-token, and domain-response averaging weight domains and response lengths. The experiments show how these choices redistribute capability gains across domains.
2. **Update sparsity and teacher overlap (Section 4).** We measure concentration of joint parameter changes, distinguishing individual steps from cumulative changes and optimizer weights from stored model weights. We formulate the teacher comparison at a shared student state and examine teacher-conditioned gradients in the appendix.
3. **Supervision density (Section 5).** We compare sampled-token and top- k supervision in single-teacher and joint training, and evaluate broader supervision at a common state. Local update concentration and online capability provide complementary measurements of these losses.

Section 6 brings these studies together in a practical training recipe, connecting supervision choices to per-domain capability retention and specifying the validation still needed.

2 PROBLEM FORMULATION

Let $\mathcal{D}_d$ be the prompt distribution for domain d , $q_d(v \mid h)$ its teacher distribution, and $p_\theta(v \mid h)$ the shared student distribution over vocabulary $\mathcal{V}$. At each update, the student samples a response y to prompt x . Its assigned teacher scores the same prefixes $h = (x, y_{<t})$. Let $\lambda_d \geq 0$ be the target domain weights, with $\sum_d \lambda_d = 1$.

For a per-position distillation loss $\ell_{d,t}$, a response-normalized objective is

$$\mathcal{L}(\theta) = \sum_d \lambda_d \mathbb{E}_{x \sim \mathcal{D}_d, y \sim p_\theta(\cdot \mid x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \ell_{d,t}(\theta) \right]. \quad (1)$$

Training differentiates the losses on the sampled responses while holding prefixes, valid-token masks, and response weights fixed. Section 3 compares the resulting batch gradient with those from other averaging rules.

At a fixed prefix, a full-vocabulary reference loss is reverse KL:

$$\ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_\theta(v) \log \frac{p_\theta(v)}{q_d(v)}. \quad (2)$$

A sampled-token surrogate, written using stop-gradient $\text{sg}[\cdot]$, is

$$\ell_{\text{PG}}(a) = -\text{sg}[\log q_d(a) - \log p_\theta(a)] \log p_\theta(a), \quad a \sim p_\theta. \quad (3)$$

At a fixed prefix, its expected gradient equals $\nabla_\theta \ell_{\text{full}}$; Appendix D gives the derivation. Vocabulary-restricted supervision instead evaluates a selected set of alternatives. We use *supervision density* to refer to the number of vocabulary alternatives scored at each prefix, and *update concentration* to describe how parameter changes are distributed across coordinates. These refer to different objects.

Table 1: Averaging rules on a fixed batch of nonempty responses. GT assigns domains their observed token shares; DT fixes domain weights; DR also gives responses equal weight within each domain.

Reduction	Batch loss
Global token (GT)	$\sum_i T_i \ell_i / \sum_i T_i$
Domain token (DT)	$\sum_d \lambda_d \sum_{i \in \mathcal{B}_d} T_i \ell_i / \sum_{i \in \mathcal{B}_d} T_i$
Domain response (DR)	$\sum_d \lambda_d \mathcal{B}_d ^{-1} \sum_{i \in \mathcal{B}_d} \ell_i$

3 TOKEN BALANCING AND RESPONSE LENGTH

3.1 HOW DOES LOSS AVERAGING CHANGE DOMAIN WEIGHTS?

The response-normalized objective in Equation 1 gives each response equal weight within its domain. Averaging all valid tokens together instead makes response length part of the weighting rule. Open-MOPD identifies this effect under token-mean aggregation (Gao et al., 2026, Section 3.3). Separating domain weights from response weights makes the effect of each averaging rule explicit.

For response i , let T_i be its valid-token count and $\ell_i = T_i^{-1} \sum_t \ell_{i,t}$ its mean token loss. Let $\mathcal{B}_d$ contain the responses from domain d , with target weights $\sum_d \lambda_d = 1$. Table 1 compares the three reductions on the same batch.

With equal prompt quotas, longer responses increase a domain’s weight under GT. DT removes this between-domain length effect, while retaining length weighting within each domain. Writing $g_i = \nabla_{\theta} \ell_i$, the exact fixed-batch relation is

$$\frac{\sum_{i \in \mathcal{B}_d} T_i g_i}{\sum_{i \in \mathcal{B}_d} T_i} = \bar{g}_d + \frac{\text{Cov}_d(T, g)}{\bar{T}_d}. \quad (4)$$

Here $\bar{g}_d$ and $\bar{T}_d$ are domain averages, and the covariance measures whether longer responses contribute systematically different gradients. DT retains this covariance term; DR removes it. Moreover, multiplying GT terms by the target domain weight divided by the observed token share gives DT exactly. This is the token-share correction in Open-MOPD (Gao et al., 2026, Section 4.1, Eq. (8)). DR makes the additional choice of weighting responses equally within each domain. Appendix B gives the derivation.

3.2 EXPERIMENT: AVERAGING RULES AND CAPABILITY BALANCE

Comparison. We train a Qwen3-1.7B student with four RL-trained teachers specializing in mathematics, code, instruction following, and science. The three branches use the same student/teacher top-64 intersection loss (I64; Appendix D) and differ in global-token (GT), domain-token (DT), or domain-response (DR) averaging. They share initialization, teachers, prompt schedule, optimizer settings, and response cap. Each update uses 16 prompts per domain; DT and DR use domain weights 1/4. Responses are sampled from each branch’s current student. We compare all three branches at update 50 on four held-out benchmarks. Appendix A gives the shared protocol and scoring.

Response length changes the loss weights. Across the first 50 GT rollout batches, mathematics receives a mean 44.05% of valid tokens and instruction following 14.98%, although each receives 25% of prompts. Under GT, these are also their mean loss weights; DT and DR assign each domain 25%. The mean mathematics weight is therefore nearly three times the mean instruction-following weight. Figure 11 gives the three branches’ token shares over the same interval.

Within-domain length weighting changes the gradient. We also hold the student and response batch fixed, then recompute I64 gradients under all three reductions. The comparison uses M-PG checkpoints 100 and 250, each with three banks of eight responses per domain. DT and DR have the same domain weights, yet the cosine between their raw gradients ranges from 0.578 to 0.911 at update 100 and from 0.399 to 0.786 at update 250. Thus response weighting changes gradient direction even after the domain weights are fixed. Table 7 reports all six banks and all three pairwise

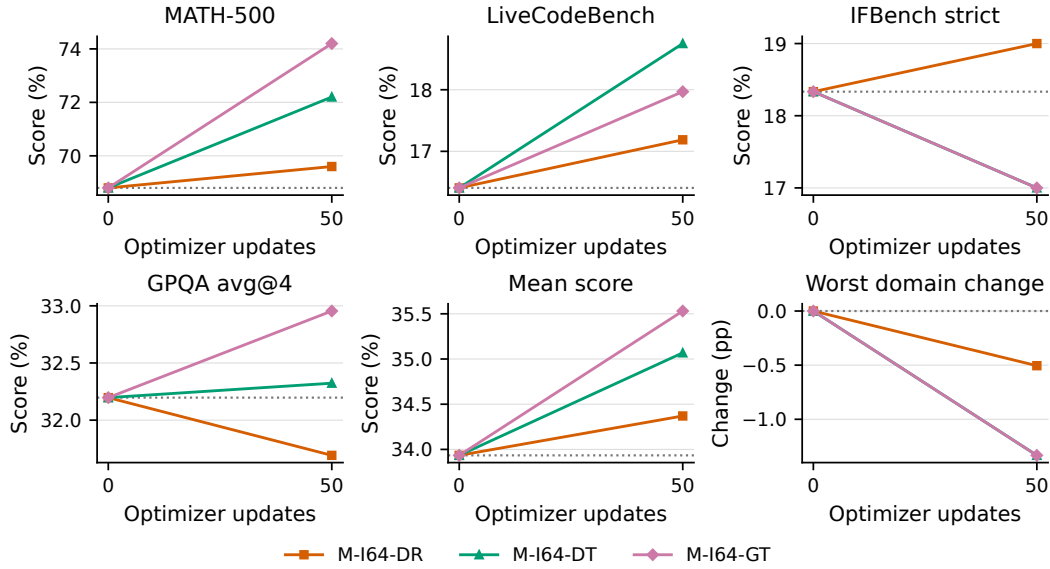


Figure 1: GT/DT/DR capability comparison at update 50. The first four panels show domain scores, followed by their arithmetic mean and the smallest domain change from initialization. The initial student is shared. Lines connect the initial and update-50 evaluations; all three branches use one common training seed.

Table 2: Capability scores (%) at update 50. Mean averages the four domain scores. Worst Δ is the smallest domain change from initialization, in percentage points.

Reduction	Math	Code	IF	GPQA	Mean	Worst Δ
DR	69.60	17.19	19.00	31.69	34.37	-0.51
DT	72.20	18.75	17.00	32.32	35.07	-1.33
GT	74.20	17.97	17.00	32.95	35.53	-1.33

comparisons. This isolates the within-batch weighting effect in Equation 4; the online branches above include the evolution of student-generated responses.

Averaging redistributes capability gains. GT achieves 74.2% on mathematics versus DR’s 69.6%, a difference of 4.6 percentage points with a paired prompt-bootstrap interval of $[1.2, 8.2]$. DR achieves 19.0% on instruction following versus 17.0% for both GT and DT. The DT-minus-DR interval on instruction following is $[-4.0, -0.33]$ points. Thus the rule with the highest mathematics score has a lower instruction-following score, while equal domain weights in DT still produce a different tradeoff from equal response weights in DR. Figure 5 reports all eight domain-level contrasts.

Table 2 gives the score comparison. GT has the highest four-domain mean (35.53%), while DR has the smallest decline on its most affected domain (-0.51 points, compared with -1.33 for DT and GT). These summaries express two different goals: average capability and preserving capability across domains. The length-weighting identity explains which training examples receive more weight; the measured scores determine the resulting tradeoff.

To complete: Later-checkpoint normalization comparison. Complete the GT, DT, and DR capability evaluations at a common later checkpoint before adding its comparison figure or table. The update-50 comparison above contains all three branches.

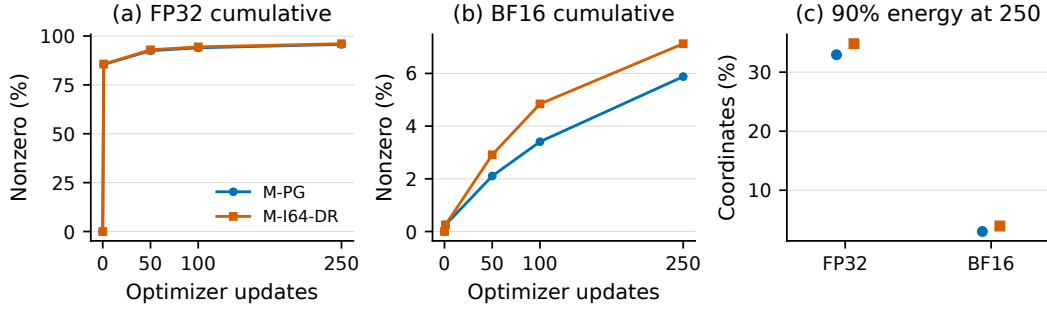


Figure 2: Cumulative changes in joint PG and I64 training through update 250. (a,b) Nonzero fractions in FP32 optimizer weights and BF16 model weights. (c) Coordinate fractions accounting for 90% of squared change at update 250. Both branches use domain-response averaging.

4 UPDATE SPARSITY AND TEACHER OVERLAP

This study proceeds from concentration of parameter changes to overlap between teachers, then to the relationship between that overlap and teacher predictions. Prior work observes concentrated changes under dense distillation supervision (Yu et al., 2026). We examine the corresponding measurements in a jointly trained student.

4.1 HOW CONCENTRATED ARE JOINT PARAMETER CHANGES?

Write $\Delta = \theta - \theta_0$ for cumulative changes from initialization and $u = \theta_{\text{after}} - \theta_{\text{before}}$ for an individual optimizer step. For either change vector z with P coordinates, we report the fraction of parameters whose absolute change exceeds threshold τ :

$$f_{\tau}(z) = \frac{1}{P} \sum_{j=1}^P \mathbf{1}\{|z_j| > \tau\}. \quad (5)$$

We also report the smallest coordinate fraction accounting for 90% of the squared change. This measures concentration without choosing an absolute threshold. Both quantities are computed separately for FP32 optimizer weights and BF16 stored model weights.

Joint-training comparison. The M-PG and M-I64-DR branches share the training settings in Appendix A, including domain-response averaging. Figure 2 compares cumulative changes at common updates 1, 50, 100, and 250. At update 250, PG changes 5.88% of BF16 coordinates, with 90% of squared change in 3.01% of coordinates. I64 changes 7.13%, with 90% of squared change in 3.94%. Both joint runs concentrate their stored parameter changes in a small fraction of the model.

The corresponding FP32 nonzero fractions are 95.73% for PG and 96.07% for I64. Their 90%-energy fractions are 32.94% and 34.82%. Thus the apparent sparsity depends strongly on which parameter representation is measured. The actual optimizer steps at update 250 change 86.35% of FP32 coordinates for PG and 86.70% for I64. Threshold curves and optimizer-state controls appear in Appendix C.

To complete: Single-teacher versus multi-teacher sparsity. Add the matching S-PG and S-I64 parameter measurements at the same checkpoints as their joint counterparts. The comparison must include individual steps and cumulative changes in both FP32 and BF16, with threshold curves and the 90%-energy fraction. State the per-domain response exposure. Populate the comparison figure only when all four configurations are available.

4.2 DO TEACHERS CHANGE THE SAME PARAMETERS IN ONE STUDENT?

Comparing teacher-specific updates requires identical copies of the joint student and its optimizer state. For teacher d , let $S_d(\rho)$ contain the largest-magnitude ρ fraction of its parameter changes. We

compare selected coordinates using Jaccard overlap:

$$J(d, e) = \frac{|S_d(\rho) \cap S_e(\rho)|}{|S_d(\rho) \cup S_e(\rho)|}. \quad (6)$$

Equal selection sizes make the teacher pairs comparable. Overlap describes locations; the signs of the changes determine whether contributions at those locations agree. This distinction follows the support-overlap analysis of Yu et al. (2026).

To complete: Teacher-specific optimizer-step overlap. Compute each teacher’s step from the same saved student and optimizer state on its routed domain. Add the complete teacher-pair overlap matrix at matched coordinate fractions, together with a common-input comparison using the same number of prefixes. The existing raw-gradient matrix belongs to the separate gradient diagnostic in Appendix C.2; it does not fill this figure.

4.3 DO DIFFERENT PREDICTIONS IMPLY DIFFERENT PARAMETER SELECTIONS?

Teacher prediction differences are measured by Jensen–Shannon (JS) divergence on the same student prefixes. We pair each teacher pair’s divergence with its parameter-selection difference, $1 - J(d, e)$. The question is whether teachers that disagree more in their predictions also change less-overlapping coordinates in the shared student. The JS definition appears in Appendix E.

To complete: Teacher distance and optimizer-step overlap. Pair the complete teacher-specific update measurements above with JS divergence at the same checkpoint and on the same prefix banks. Show all teacher pairs using consistent coordinate selection sizes. Add the scatter plot and its interpretation together after this pairing is complete. The separate gradient comparison is retained in Appendix E.

5 DOES SUPERVISION DENSITY CHANGE UPDATE SPARSITY?

Broader vocabulary supervision can change the gradient at a prefix without proportionally increasing the number of parameters changed by the optimizer. Dense supervision and concentrated checkpoint changes coexist in the experiments of Yu et al. (2026). We examine this relationship through a local comparison at a common state and online PG/I64 comparisons in single-teacher and joint training.

5.1 COMPARISON AND MEASUREMENTS

PG uses the sampled-token loss in Equation 3. I64 scores the intersection of the rollout student’s and teacher’s top-64 sets, with detached student weights normalized over the original student set. Its formula appears in Appendix D. The online configurations are S-PG and S-I64 with the mathematics teacher, and M-PG and M-I64-DR with four routed teachers. Within each setting, the branches share initialization, teachers, domain-response averaging, optimizer settings, and prompt schedule. The single-teacher branches use 64 mathematics responses per update; joint branches use 16 per domain. We compare PG and I64 within each setting at common checkpoints.

The local experiment evaluates PG, I64, corrected teacher-top-64 supervision (T64), and full-vocabulary reverse KL on the same four prefix banks at M-PG update 100. Each bank contains one response per domain and six selected prefixes per response, for 16 responses and 96 prefixes in total. Responses use the training cap of 4,096 tokens; two responses reach the cap. Every loss uses identical prefix weights and the same saved optimizer state. We compute the fraction of changed parameters and the fraction of squared step norm carried by the largest 1% of coordinates.

5.2 UPDATE CONCENTRATION

Local changes are similarly concentrated across losses. Figure 3 shows that all four losses change about 88.0% of FP32 coordinates. Mean BF16 nonzero fractions are 0.149% for PG and 0.150% for I64, T64, and full vocabulary. The largest 1% of coordinates carry about 8.8% of the FP32 squared step norm and all of the BF16 squared step norm. Local concentration is therefore very similar across the four losses at this shared state. The shared Adam history contributes to this result: a zero-current-gradient control alone changes 0.140% of BF16 coordinates.

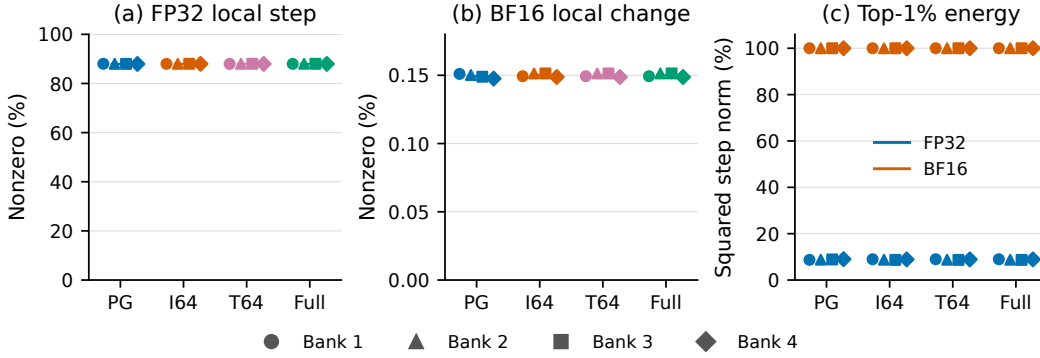


Figure 3: Local supervision comparison at M-PG update 100. (a,b) Nonzero fractions for FP32 optimizer steps and BF16 model changes. (c) Squared step norm in the largest 1% of coordinates. Points show four prefix banks; every loss uses the same saved optimizer state.

The teacher’s top-64 tokens contain a mean 99.918% of student probability across these prefixes, with a minimum of 96.709% at an individual prefix. The selected vocabulary therefore retains most of the student probability in this experiment. Appendix D gives the sampling and coverage details; Appendix C provides separate threshold and optimizer-state controls.

Joint trajectories accumulate different changes. Figure 2 compares cumulative PG and I64 changes under the same joint training protocol. At update 250, I64 changes 7.13% of BF16 coordinates versus 5.88% for PG; the coordinate fractions accounting for 90% of squared change are 3.94% and 3.01%. Broader vocabulary supervision therefore accompanies broader cumulative activity in this comparison, while both remain concentrated. The local experiment fixes the student, prefixes, and optimizer state; the online comparison includes the feedback between each loss and the responses its student generates.

5.3 CAPABILITY UNDER SINGLE-TEACHER AND JOINT TRAINING

Table 3 reports PG/I64 pairs: updates 100, 250, and 500 for the single-teacher branches and 50, 100, and 250 for the joint branches. In single-teacher training, PG’s mathematics score exceeds I64 by 2.8 points at update 100, whereas I64 exceeds PG by 2.2 points at update 500. At update 500, I64 also scores 0.67 points higher on instruction following, while PG scores 1.56 points higher on code and 1.77 points higher on GPQA.

In joint training, both losses reach 26.0% on instruction following at update 250. I64 reaches 74.8% on mathematics, 17.97% on code, and 33.96% on GPQA, compared with PG’s 73.4%, 16.41%, and 30.18%. The I64-minus-PG GPQA difference is 3.79 points, with a paired prompt-bootstrap interval of $[0.00, 7.58]$. Its four-domain mean is 38.18%, compared with 36.50% for PG. These endpoint scores favor I64 in the joint setting, while the single-teacher comparison retains domain tradeoffs. Figure 4 shows all matched trajectories, and Table 6 gives all eight endpoint differences with their paired intervals.

The capability comparison and the local update comparison answer different parts of the supervision question. Broader feedback yields almost unchanged concentration at the fixed state, but the trained students develop different capability profiles. The matching single-teacher parameter measurements are specified in the annotation in Section 4.

6 TOWARD A STABLE MULTI-TASK OPD RECIPE

The three studies suggest that capability retention should guide the choice of averaging rule, teacher feedback, and training checkpoint. A useful starting point is to specify the acceptable tradeoff before training. Let $s_d(\theta)$ be a development-set score for domain d , oriented so that higher is better, and let $\epsilon_d \geq 0$ be a prespecified tolerance relative to the initial student. The proposed recipe selects

Table 3: Online PG/I64 capability pairs (%). Each pair shares the setting and update count. Mean averages the four domain scores. Single-teacher and joint updates have different per-domain response counts.

Configuration	Updates	Math	Code	IF	GPQA	Mean
S-PG	100	77.40	18.75	18.67	31.82	36.66
S-I64	100	74.60	18.75	17.67	33.21	36.06
S-PG	250	73.80	20.31	17.00	35.35	36.62
S-I64	250	74.40	21.09	18.33	32.45	36.57
S-PG	500	73.80	19.53	18.00	31.82	35.79
S-I64	500	76.00	17.97	18.67	30.05	35.67
M-PG	50	69.60	16.41	18.67	28.03	33.18
M-I64-DR	50	69.60	17.19	19.00	31.69	34.37
M-PG	100	72.00	17.19	19.33	31.19	34.93
M-I64-DR	100	71.60	17.97	19.00	32.70	35.32
M-PG	250	73.40	16.41	26.00	30.18	36.50
M-I64-DR	250	74.80	17.97	26.00	33.96	38.18

checkpoints subject to

$$s_d(\theta) \geq s_d(\theta_0) - \epsilon_d \quad \text{for every domain } d. \quad (7)$$

Among checkpoints meeting these conditions, the proposed rule selects the one with the highest prespecified aggregate utility, retaining the initial student if none improves it. Selection should account for score uncertainty at the chosen tolerances, followed by evaluation on a separate test set. The constraints govern checkpoint selection rather than add differentiable terms to the distillation loss.

With the retention goal fixed, the averaging rule should express the intended importance of each training response. For an objective that weights responses equally within each domain, we use DR with explicit domain weights λ_d . In our four-domain setting, equal weights and 16 prompts per domain provide the starting allocation. If longer responses are deliberately assigned more weight within a domain, DT expresses that alternative. Under either reduction, the prompt quota and the domain loss weight remain separate choices: with every domain represented and its loss averaged separately, increasing a quota at fixed λ_d changes exposure and estimation variance without directly increasing that domain’s expected gradient weight. Generated lengths and truncation rates should therefore be recorded alongside both quantities when diagnosing an imbalance.

Once these weights are explicit, vocabulary feedback can be chosen using its capability profile and computational cost. The joint comparison in Section 5 motivates I64 as a candidate alongside DR, with PG retained as a reference. Its feedback scale also deserves attention. For the student candidate set S and teacher intersection I in Equation 9, the retained student weights sum to $r(h) = p_\theta(I \mid h)/p_\theta(S \mid h)$. At a fixed prefix with a nonempty intersection and detached coefficients, the I64 gradient is $r(h)$ times the gradient obtained by renormalizing the same weights over I . Thus removing length weighting does not remove variation in feedback scale across prefixes. Monitoring this retained mass and empty intersections by domain helps assess whether a fixed candidate size supplies comparable coverage; increasing or renormalizing a small intersection requires validation because it can also amplify unreliable feedback.

If a domain repeatedly falls below its retention tolerance, the response should depend on the diagnosed cause. Exposure, length, and coverage checks can reveal inadequate sampling or unintended weighting. When these do not explain the decline, teacher interactions should be examined at a common student and optimizer state. The overlap diagnostic defined in Section 4 describes shared update locations; a further functional check is to apply each teacher’s proposed step and measure the change in fixed probe losses from all domains. A zero-current-gradient step provides a reference for the contribution of optimizer history. This comparison can identify local cross-domain harm that coordinate overlap alone does not establish, while subsequent capability evaluations determine whether it persists along training. Gradient modification or task-specific parameters become interventions to test when such harm remains reproducible after the allocation and coverage checks.

The default recipe keeps domain weights and prompt quotas fixed; any adaptive variant should state its trigger and adjustment rule separately. To maintain previous-domain exposure without changing

the on-policy protocol, retain prompts from those domains and generate fresh responses with the current student before teacher scoring. The resulting starting recipe combines explicit response-level weighting, I64 feedback with coverage monitoring, and retention-based checkpoint selection. Its stability must be assessed both along the training trajectory and across independent runs: an acceptable endpoint does not establish that earlier checkpoints avoided degradation.

To complete: Recipe validation. Complete a matched GT/DR $\times$ PG/I64 comparison through update 250 with at least three paired training seeds, adding PG-GT to the existing configurations. Retain I64-DT to separate between-domain and within-domain length effects. Match per-domain prompt exposure and update counts, report token and compute costs separately, and evaluate domain trajectories, worst-domain changes, and retention across seeds. Use an independent development set for checkpoint selection. Complete the common-state teacher-step comparisons with cross-domain probe losses and a zero-current-gradient reference, and audit I64 retained mass by domain. These comparisons are pending; no aggregate recipe result is reported here.

7 RELATED WORK

MOPD combines specialized teachers through feedback on student-generated responses (Ma et al., 2026). Open-MOPD studies capability imbalance and introduces token-share balancing, gap-based weighting, and reward refresh (Gao et al., 2026). Its token-share correction maps directly to domain-token averaging under the same batch and target domain weights. Our first study separates this between-domain correction from within-domain response weighting and compares their capability outcomes.

Mukherjee et al. (2025) investigate small parameter subsets that support reinforcement-learning fine-tuning. Yu et al. (2026) find concentrated checkpoint changes under dense distillation supervision and overlap between the changes produced by different teachers in separately trained students. Our second question concerns teachers acting from the same joint student and optimizer state. Our third compares vocabulary supervision through both parameter concentration and capability. Shen et al. (2026) study OPD parameter geometry and select response positions for learning; their position selection concerns a different supervision dimension from the vocabulary alternatives scored at each prefix here.

8 DISCUSSION

The experiments cover one model family and one training seed. The local supervision comparison uses four prefix banks with high top- k probability coverage, limiting what it establishes about broader vocabulary tails or other student states. The bootstrap intervals quantify evaluation-question uncertainty at fixed checkpoints, not variability across training runs. These limits apply to the proposed recipe as well as to the individual comparisons.

Local changes in a distillation loss and long-term capability transfer remain distinct measurements. Student-generated responses evolve during training, so a fixed-state diagnostic does not capture the full trajectory. The recipe in Section 6 uses such diagnostics to guide decisions, with stability left to prospective tests of domain retention under matched budgets and repeated training runs.

REPRODUCIBILITY STATEMENT

The accompanying evidence bundle contains the values used in every figure and table, source hashes, configuration records, a scoring audit, and scripts that regenerate the plots on CPU. Twelve capability suites are verified against response-level artifacts; four single-teacher suites use previously verified summaries from a second node. Appendix A gives the evaluation protocol, checkpoint coverage, and uncertainty procedure. Appendix C specifies the parameter coordinates and local optimizer calculations.

ETHICS STATEMENT

The experiments use public model families and benchmark datasets to study optimization and capability transfer. We report standard benchmark scores with the stated evaluation denominators. Generative AI assisted manuscript editing and analysis code; the authors are responsible for the reported measurements and their interpretation.

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A EXPERIMENTAL PROTOCOL AND CAPABILITY RESULTS

The six configurations are shared across the three questions. The three I64 reductions address token balancing (Section 3); single-teacher and joint PG/I64 branches define the update-sparsity comparison (Section 4) and the supervision-density comparison (Section 5). Each reported comparison uses common completed checkpoints for its branches. The source inventory additionally records completed evaluations outside those comparisons.

A.1 MODEL, DATA, AND TRAINING PROTOCOL

The student is Qwen3-1.7B (Yang et al., 2025), revision 70d244cc86cc (the full revision and weight hashes are in the frozen protocol). The four frozen teachers are mathematics, code, instruction-following, and science RL descendants of the same initial model. Student and teacher use the native chat template with thinking disabled, the same assistant prefix, and no suffix removal. Generation stops at token 151643 or 151645; the model EOS token is 151645. The vocabulary contains 151,936 entries.

All six configurations use AdamW with learning rate 2.5×10^{-7} , $(\beta_1, \beta_2) = (0.9, 0.98)$, $\epsilon = 10^{-8}$, zero weight decay, constant learning rate, and global gradient clipping at norm 1. Training sampling uses temperature 1 and top- $p = 1$, with a 2,048-token prompt cap and 4,096-token response cap. One fresh response is sampled per prompt. Training uses seed 42. An optimizer update aggregates 64 responses in sixteen four-response slices, with one response per execution microbatch. The optimizer is applied once after all domain blocks are accumulated. Consequently, the scalar logger’s `train/step` counts gradient slices; all checkpoint axes in this paper count actual optimizer updates.

The multi-teacher configurations allocate 16 prompts to each domain per update and use target domain weights $1/4$. Single-teacher configurations allocate all 64 prompts to mathematics. Prompts follow a seeded shuffled schedule, repeated after exhaustion. The training splits contain 21,928 mathematics, 19,041 code, 16,447 instruction-following, and 19,542 science records from the frozen four-domain prompt bundle. Each domain has separate 64-record diagnostic and held-out splits. The frozen protocol records their exact hashes and the teacher conversion manifests. I64 starts from initialization with the intersection objective in Equation 9.

To complete: Training data provenance. Add the upstream dataset name, release or revision, and preprocessing reference for each of the four training splits. The frozen protocol identifies the exact files and sizes; the manuscript also needs a reader-facing mapping to their sources.

A.2 EVALUATION AND SCORING

Each capability suite contains MATH-500 (500 questions from the MATH benchmark (Hendrycks et al., 2021); dataset release), a fixed 128-question LiveCodeBench v6 subset (Jain et al., 2024), IFBench strict (300 questions) (Pyatkin et al., 2025), and GPQA Diamond (198 questions) (Rein et al., 2023). Mathematics, code, and instruction following use one greedy response per question. GPQA uses four responses per question at temperature 1 and top- $p = 1$. Its score is the mean correctness over all four responses, then over questions: average@4 accuracy. Evaluation permits up to 32,768 response tokens, bounded by the remaining configured 32,768-token context. Each complete suite contains 1,720 responses.

The inventory contains sixteen complete suites totaling 27,520 responses. Twelve suites have local response-level artifacts: the shared initial student, nine multi-teacher checkpoints, and both single-teacher endpoints at update 500. Their 20,640 response records were checked against artifact hashes, expected sample counts, unique prompt/sample indices, and logged aggregate scores. The code artifacts report zero sandbox infrastructure errors. Recomputing all 9,504 GPQA responses with the recorded `final-answer-v2` scorer gives zero score changes. Four single-teacher suites contribute 6,880 responses through previously verified completion summaries from a second node. Their per-question records remain on that node. Tables and comparison figures retain only checkpoints shared by all branches in each comparison; other completed scores remain in the inventory.

Table 4: Capability scores (%) at complete comparison checkpoints. The initial student is shared. GT/DT/DR use update 50; PG/I64 use common updates within each teacher setting. Scores use all evaluated responses.

Configuration	Updates	Math	Code	IF	GPQA
Initial	0	68.80	16.41	18.33	32.20
S-PG	100	77.40	18.75	18.67	31.82
S-PG	250	73.80	20.31	17.00	35.35
S-PG	500	73.80	19.53	18.00	31.82
S-I64	100	74.60	18.75	17.67	33.21
S-I64	250	74.40	21.09	18.33	32.45
S-I64	500	76.00	17.97	18.67	30.05
M-PG	50	69.60	16.41	18.67	28.03
M-PG	100	72.00	17.19	19.33	31.19
M-PG	250	73.40	16.41	26.00	30.18
M-I64-DR	50	69.60	17.19	19.00	31.69
M-I64-DR	100	71.60	17.97	19.00	32.70
M-I64-DR	250	74.80	17.97	26.00	33.96
M-I64-DT	50	72.20	18.75	17.00	32.32
M-I64-GT	50	74.20	17.97	17.00	32.95

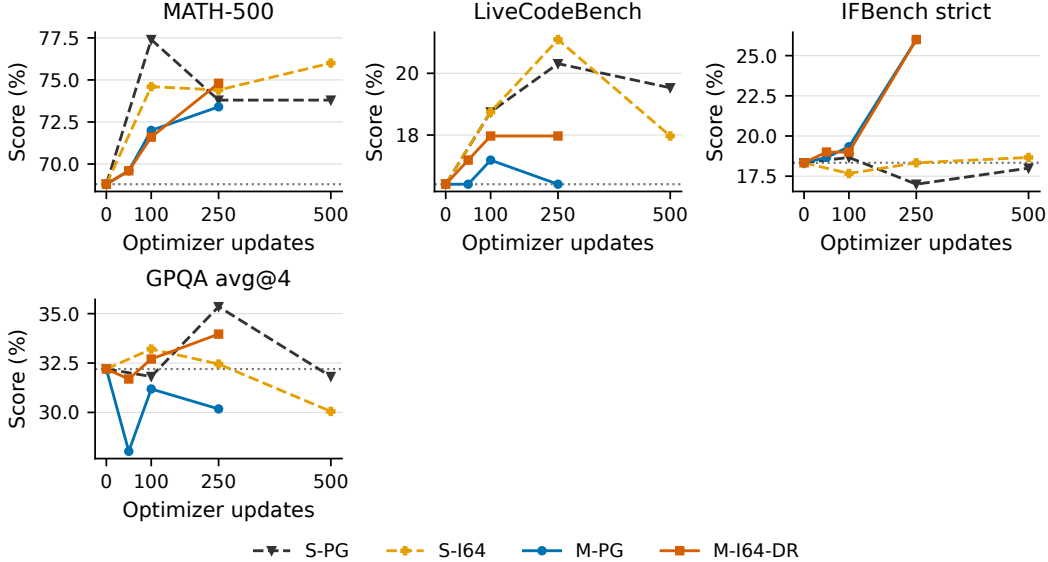


Figure 4: PG/I64 capability comparisons from the shared initial student. M denotes four routed teachers and S the mathematics teacher. Lines connect common measured checkpoints: 100/250/500 for S and 50/100/250 for M. Single-teacher updates contain four times as many mathematics responses as joint updates.

The four RL teachers were evaluated separately on their respective domain benchmarks with the compatible native prompt and decoding protocol (Table 5). The science teacher’s 792 saved responses also agree exactly with the current GPQA scorer. These teacher scores describe the available supervision on each target domain.

A.3 PAIRED COMPARISONS AND EVIDENCE PROVENANCE

For the paired comparisons, question identity includes the prompt, label, and question metadata, with post-generation response and sandbox diagnostics removed. Correspondence is verified before pairing. We bootstrap the per-question score differences 10,000 times using seed 1042; each GPQA question retains its four-response cluster. Figure 5 shows percentile 95% intervals. These

Table 5: Domain teacher reference evaluations, retained from the compatible September 6 evaluation campaign. Science uses GPQA average@4.

Teacher / target domain	Responses	Score (%)	Truncation (%)
Math	500	75.60	6.80
Code	128	23.44	24.22
If	300	29.00	7.33
Science	792	34.60	0.51

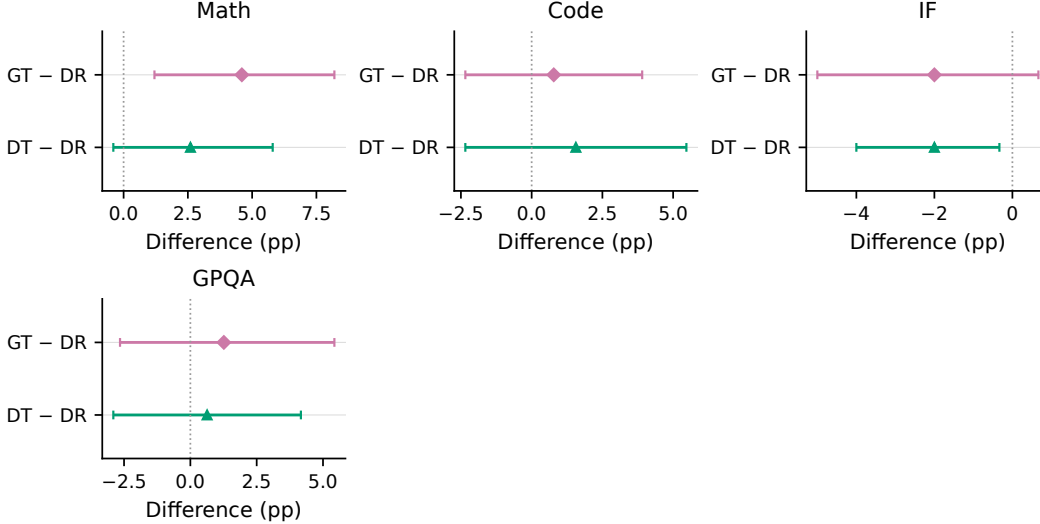


Figure 5: All paired update-50 I64 reduction contrasts against DR. Points give score differences in percentage points and bars give nominal 95% paired prompt-bootstrap intervals. GPQA resamples questions with their four responses together.

are nominal, unadjusted intervals over evaluation questions, conditional on the trained checkpoints. Training-seed variation is a separate source of uncertainty. The same procedure supplies the initial-to-checkpoint comparisons in the evidence bundle.

The source bundle `experiments/aligned_evidence_20260910` freezes the plotted measurements, prompt-level score records, source SHA-256 hashes, scoring implementation, protocol, and local diagnostic code. Figure generation reads only this bundle and runs on CPU. The manifest identifies the source artifact for each measurement. The aligned series uses the common initial parent and native response semantics throughout; earlier Base initialization, Student64-to-I64 continuations, and superseded teacher-top- k runs are outside this series.

B LENGTH-WEIGHTING DERIVATION

Fix a rollout batch, its valid-token masks, and domain weights λ_d . Any PG coefficients are detached. For $n_d = |\mathcal{B}_d|$, let $\bar{T}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} T_i$ and $\bar{g}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} g_i$. Define the empirical covariance using the same $1/n_d$ normalization:

$$\text{Cov}_d(T, g) = \frac{1}{n_d} \sum_{i \in \mathcal{B}_d} (T_i - \bar{T}_d)(g_i - \bar{g}_d). \quad (8)$$

Expanding gives $n_d^{-1} \sum_i T_i g_i - \bar{T}_d \bar{g}_d$. Dividing by $\bar{T}_d$ yields Equation 4. GT combines the resulting domain gradients using observed token shares; DT combines them using λ_d ; DR combines $\bar{g}_d$ using λ_d . Multiplying each domain’s globally averaged terms by its target weight divided by its observed token share therefore gives DT exactly.

Table 6: All PG/I64 domain contrasts at the last common capability checkpoints: single-teacher update 500 and joint update 250. Differences are I64 minus PG in percentage points. Intervals are nominal 95% paired question-bootstrap intervals conditional on the trained checkpoints.

Setting	Update	Domain	I64 – PG (pp)	95% interval (pp)
Single	500	Math	+2.20	[-0.80, +5.00]
Single	500	Code	-1.56	[-3.91, 0.00]
Single	500	IF	+0.67	[-1.67, +3.00]
Single	500	GPQA	-1.77	[-5.30, +1.77]
Joint	250	Math	+1.40	[-1.20, +4.00]
Joint	250	Code	+1.56	[0.00, +3.91]
Joint	250	IF	0.00	[-2.67, +2.67]
Joint	250	GPQA	+3.79	[0.00, +7.58]

Table 7: Fixed-batch averaging comparison. Entries are raw-gradient cosines; each row uses one shared student and response bank. DR and DT share domain weights and differ only in within-domain response weighting.

Updates	Batch	DR–DT	DR–GT	DT–GT
100	1	0.7838	0.6219	0.9266
100	2	0.9110	0.5180	0.7010
100	3	0.5780	0.6659	0.9745
250	1	0.6240	0.4072	0.8742
250	2	0.7864	0.3302	0.4953
250	3	0.3990	0.3569	0.9612

The derivation is a finite-batch weighted-average identity. Response sampling, masks, and lengths are held fixed. Its population analogue involves an expectation of a ratio; replacing that expectation by a ratio of expectations requires additional assumptions or a large-batch limit.

B.1 FIXED-BATCH GRADIENT COMPARISON

At M-PG checkpoints 100 and 250, three banks (seeds 1042, 1043, and 1044) each contain eight responses per domain. Responses use the training prompt format and 4,096-token response cap. For each bank, we compute each response’s mean I64 gradient once at the saved BF16 model values using FP32 computation, then aggregate those gradients under DR, DT, and GT. All three reductions use the same responses, token masks, teacher scores, and model coordinates. Student and optimizer snapshots match within each checkpoint.

Table 7 reports every pairwise raw-gradient cosine. The domain-wise covariance decomposition is checked in FP64 over all 24 bank-domain groups; the largest relative residual is 1.53×10^{-15} . This numerical check verifies the aggregation identity. The observed gradient directions establish that the covariance contribution is nonzero on these batches.

C PARAMETER AND OPTIMIZER MEASUREMENTS

C.1 COORDINATES AND RECORDED QUANTITIES

The online record distinguishes raw aggregate gradients, clipped gradients, actual FP32 optimizer steps, cumulative FP32 master changes, and cumulative BF16 model changes. Cumulative changes use the same initialization reference. All global statistics use unique model coordinates, excluding vocabulary padding and shared duplicates. Per-layer statistics and absolute-threshold sensitivities are retained in the frozen measurements. The sparsity value is the fraction at or below a threshold; figures report its complement. Energy concentration uses squared Euclidean norm and ranks coordinates by absolute change.

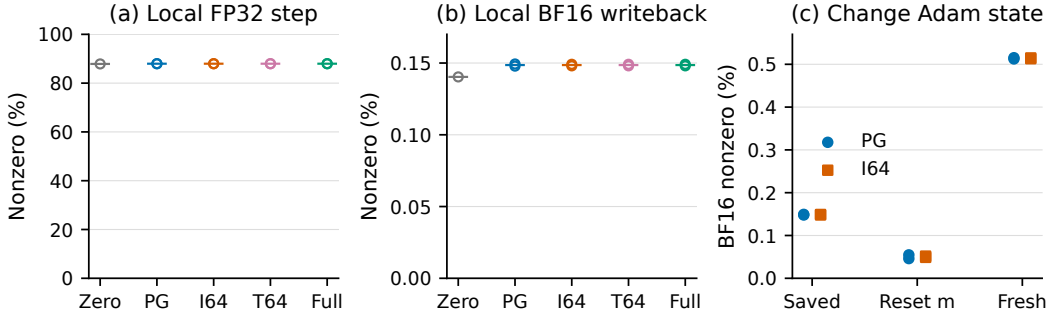


Figure 6: Supporting optimizer-state controls at M-PG update 100. Points show two response banks; the zero-gradient control is evaluated once. (a,b) Nonzero fractions for FP32 steps and BF16 changes. (c) Saved moments, reset first moment, and fresh moments and clock. Weight decay is zero.

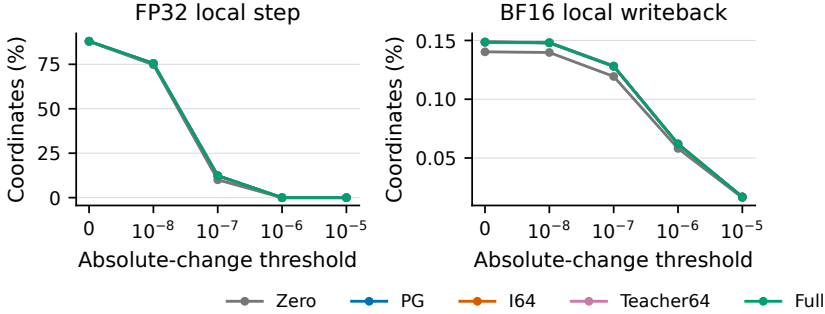


Figure 7: Absolute-threshold sensitivity of the saved-state local steps. Lines show means over two banks and shading shows their range; the zero-gradient control has one value. The two panels use the same thresholds and separate FP32 master changes from BF16 writeback.

For local calculations, the exported checkpoint contains FP32 master values, BF16 model values, first and second moments, optimizer hyperparameters, and step clocks. The BF16 values match all 1,720,574,976 coordinates in the native exported model. Teacher weight values are verified in the same precision. Local forward/backward computations evaluate these values in FP32 with TF32 disabled. All proposed steps are computed from the same saved state with a non-fused Adam implementation. The online training optimizer uses its own fused implementation.

We evaluate three local states. *Saved* retains both moments and the step clock. *Reset m* zeros the first moment while retaining the second moment and clock. *Fresh* zeros both moments and resets the clock. Global norm clipping at 1 is applied before the Adam calculation, and weight decay is zero. For the zero-gradient branch, the saved first moment continues to decay and move the master weights. A no-step master-to-BF16 cast gives an exact match to the saved model. Thirteen numerical checks cover the Adam formulas against PyTorch, loss gradients, parameter-coordinate inner products, support ties, and zero-vector handling. Figure 7 shows how absolute thresholds change the measured local activity.

The zero-current-gradient branch changes 0.1403% of BF16 coordinates, compared with 0.1486% for PG. Resetting the first moment gives 0.0504% for PG; resetting both moments and the clock gives 0.5140%. These controls explain why a sparse gradient selection cannot by itself identify the parameters changed by an optimizer step. They supplement the update-sparsity measurements in Sections 4 and 5.

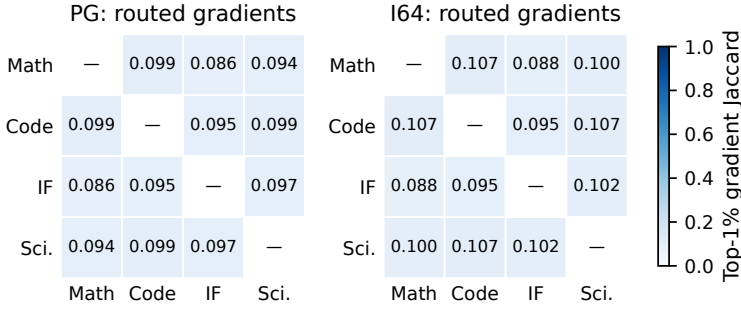


Figure 8: Teacher-pair raw-gradient comparison at M-PG update 100. Entries show mean Jaccard over two banks for the largest-magnitude 1% of routed gradient coordinates. Both panels use the same scale.

C.2 RESPONSE BANKS, OVERLAPS, AND GRADIENT MAGNITUDES

Banks use seeds 42 and 43, one response per domain per bank, a 2,048-token prompt cap, and a 256-token generation cap. All eight responses reach the cap. Four response positions are sampled without replacement per response. Each row’s four prefixes receive equal weight. For routed gradients, each teacher scores its domain row; for common-bank gradients, it scores all four rows with equal domain weights. Crossed comparisons score every domain row with every teacher, keeping the number of prefixes per gradient at four. The 192 crossed-pair records and 48 routed/common pair records share underlying gradients, so points represent paired conditions within banks.

Top- k coordinate sets rank absolute gradient values, using a stable coordinate index to break cutoff ties. The diagnostic records the cutoff and whether a tie crosses the selection boundary. These tie flags accompany the source metrics, including overlaps from nearly degenerate gradients. Table 8 supplies the routed PG norms; the first bank’s instruction-following gradient has norm 1.53×10^{-9} . Cosine normalizes magnitude, so opposite directions for such a row carry very little absolute gradient mass. Jaccard overlap measures selected locations and cosine measures signed direction. For reference, independent uniform sets of fraction ρ have expected-intersection to expected-union ratio $\rho/(2-\rho)$, or approximately 0.00503 at $\rho = 1\%$. This reference uses uniform coordinates; layer and weight-magnitude structure remain in the measured gradients.

Applying Equation 6 to the largest-magnitude 1% of raw-gradient coordinates gives the six-pair comparison in Figure 8. Across the two banks, routed gradients have mean Jaccard overlap 0.0951 for PG and 0.0999 for I64. Their ranges are 0.0744–0.1077 and 0.0737–0.1178. When every teacher instead scores all four domain responses, the respective means are 0.3808 and 0.4577. The latter comparison changes both input composition and prefix count; the crossed input control below fixes the count.

Holding the teacher fixed while changing the input domain yields mean gradient cosines of 0.0059 for PG and 0.0072 for I64. Holding the input domain fixed while changing teachers produces both aligned and opposed gradients. For routed gradients the mean cosines are -0.0020 and 0.0081 , respectively, despite partial coordinate overlap. Figure 9 supplies this input and direction context for the teacher-overlap results in Section 4.

D VOCABULARY SUPERVISION AND COVERAGE

Four-bank local experiment. The local supervision comparison in Figure 3 uses four response banks with seeds 2042–2045 at M-PG update 100. Each bank contains one response per domain. Generation uses the training prompt format, a 2,048-token prompt cap, and a 4,096-token response cap. The 16 response lengths range from 81 to 4,096 tokens; two responses reach the cap. Six positions per response give 96 prefixes. PG actions are freshly sampled from the student at the cached prefixes. Every loss uses the same prefixes and domain-response weights. Gradients are globally clipped at norm 1 before applying the same saved Adam state.

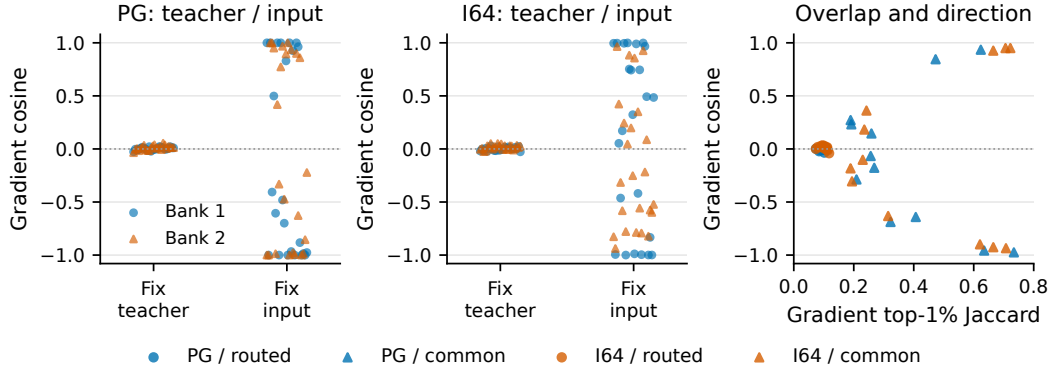


Figure 9: Supporting input controls for teacher overlap at M-PG update 100. Left and center: crossed comparisons keep four prefixes per gradient. Right: top-1% gradient Jaccard versus signed cosine for routed and common-bank gradients. Dots share teachers and banks and are dependent comparisons.

The teacher-top-64 student probability mass averages 99.918% across the 96 prefixes. The bank-domain averages range from 99.279% to 100%, and the lowest individual-prefix mass is 96.709%. The source bundle records all 16 bank-domain coverage entries and every proposed update. Student BF16 values match the saved model exactly. The local loss and Adam implementations are the same as in the two-bank controls in Appendix C.

Implemented intersection loss. Let S be the rollout student’s top-64 set, T the teacher’s top-64 set, and $I = S \cap T$. The online I64 loss is

$$\ell_{I64} = - \sum_{v \in I} \text{sg} \left[\frac{p_{\theta}(v)}{\sum_{u \in S} p_{\theta}(u)} (\log q_d(v) - \log p_{\theta}(v)) \right] \log p_{\theta}(v). \quad (9)$$

All probabilities use full-vocabulary normalization. Student weights are normalized over S before masking to I ; current student probabilities are recomputed at the saved candidate positions. An empty intersection contributes zero. PG uses detached log-ratio coefficients without advantage or policy-ratio clipping. I64 and PG are the online objectives used throughout the comparisons.

Teacher-top-64 local reference. The corrected teacher-top-64 diagnostic uses the fixed teacher set T and full-vocabulary probabilities:

$$\ell_{T64} = \sum_{v \in T} \left[p_{\theta}(v) \log \frac{p_{\theta}(v)}{q_d(v)} - p_{\theta}(v) + q_d(v) \right]. \quad (10)$$

Its gradient is $\sum_{v \in T} p_{\theta}(v) \log(p_{\theta}(v)/q_d(v)) \nabla_{\theta} \log p_{\theta}(v)$. I64 instead uses the detached weights in Equation 9, normalized over the original student set S . T64 and full vocabulary are local objective comparisons at the M-PG checkpoint; online capability curves compare PG and I64.

For a fixed prefix, differentiating Equation 2 gives

$$\nabla_{\theta} \ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_{\theta}(v) \left(\log \frac{p_{\theta}(v)}{q_d(v)} + 1 \right) \nabla_{\theta} \log p_{\theta}(v) \quad (11)$$

$$= \mathbb{E}_{a \sim p_{\theta}} \left[\log \frac{p_{\theta}(a)}{q_d(a)} \nabla_{\theta} \log p_{\theta}(a) \right], \quad (12)$$

using $\sum_v \nabla_{\theta} p_{\theta}(v) = 0$. The identity conditions on prefixes and fixed weights. Original rollout losses additionally use the realized response and its eventual length-dependent weight. Table 8 reports the corresponding coverage for the two-bank controls used to interpret gradient overlap and optimizer history.

Table 8: Coverage and routed gradient scale for the separate two-bank gradient and optimizer controls (256-token generation cap). Probability mass and full-vocabulary student entropy (nats) are averaged over each row’s four prefixes. A displayed mass of 100% may reflect rounding at the reported precision.

Bank	Domain	Mass in T (%)	Mass in I (%)	Entropy	$\ g_{PG}\ _2$
1	Math	100.0000	100.0000	0.3243	132.6
1	Code	99.9968	99.9965	0.7701	163.8
1	IF	100.0000	100.0000	1.85×10^{-5}	6.13×10^{-9}
1	Science	99.9184	99.9162	0.7405	54.82
2	Math	99.9991	99.9991	0.3221	2.605
2	Code	100.0000	100.0000	0.2962	3.167
2	IF	100.0000	100.0000	0.5004	55.52
2	Science	99.9993	99.9992	0.2442	0.8505

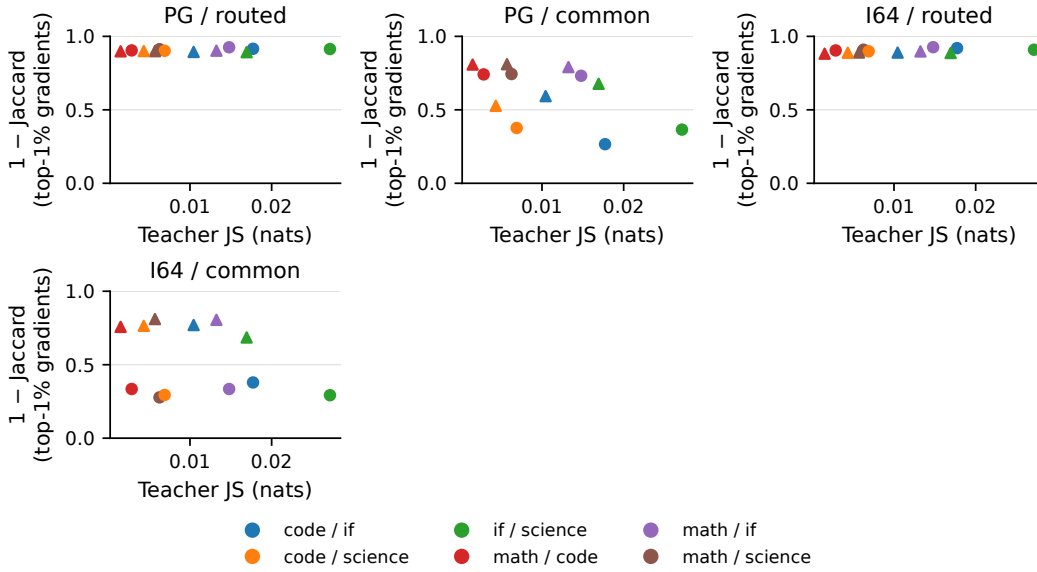


Figure 10: Teacher prediction differences and raw-gradient selection differences at M-PG update 100. Colors identify the six teacher pairs; circles and triangles identify the two banks. Each panel pairs the same teacher JS values with PG or I64 gradient overlap on routed or common inputs.

E TEACHER DISTRIBUTION DISTANCE

Teacher distance is computed on the full next-token vocabulary at matched prefixes:

$$JS(q_d, q_e) = \frac{1}{2} D_{KL}(q_d \| m) + \frac{1}{2} D_{KL}(q_e \| m), \quad m = \frac{1}{2}(q_d + q_e). \quad (13)$$

We average over the same equally weighted domains and prefixes for every pair. Figure 10 pairs all twelve bank-pair JS measurements with their corresponding gradient overlaps. Teacher JS ranges from 0.00151 to 0.02714 nats. Routed gradient-selection differences lie in a narrow band, while common-input differences vary more widely. The ordering of teacher pairs changes with the input bank and supervision choice. These pairwise observations share teachers, prefixes, and a student checkpoint.

F TOKEN SHARES UNDER THE THREE AVERAGING RULES

Figure 11 supports the loss-weighting comparison in Section 3. Each branch allocates 25% of prompts to every domain. Under GT, the plotted token shares also give the domain loss weights. DT and DR fix those weights to 25% even when generated token shares differ. Records at the

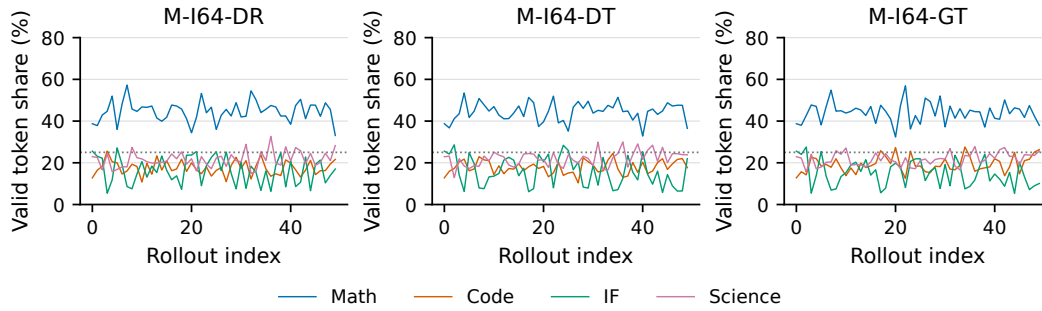


Figure 11: Valid-token shares over the first 50 rollout batches for mathematics, code, instruction following, and science under the three I64 reductions. The dotted reference is the common 25% prompt share. Rollout index r precedes optimizer update $r + 1$. GT uses these shares as domain loss weights; DT and DR use fixed domain weights.

same rollout clock are merged across metric writes before plotting; the plot shows the common first 50 rollout batches without smoothing, matching the capability comparison in Section 3. Longer histories remain in the source bundle.